NAD+ (Nicotinamide Adenine Dinucleotide)

Step-by-Step Reconstitution Process

- Gather supplies: BAC water, alcohol swabs, insulin syringe, vial
- Removing plastic cap. Carefully pop off the plastic cap, do not remove metal flange.
- Clean both vials: Wipe the rubber stoppers with alcohol swabs
- Draw BAC water: Pull your chosen amount into the syringe
- Inject slowly: Add the water to the peptide vial, aiming it at the glass wall—not directly onto the powder
- Swirl gently: Roll the vial between your palms. Never shake.
- Wait: Let it sit 5-10 minutes. Solution should be crystal clear.
- Store: Refrigerate immediately. Good for 28-30 days once reconstituted.

If it's cloudy or won't dissolve: Let it sit longer. Try gentle warming (under your arm or in lukewarm water). If still cloudy after 30 minutes, contact your source.

Dosing Protocol

Understanding the Different Approaches

Unlike some peptides that only work when dysfunction is present, NAD+ decline affects everyone with age. By your 40s, NAD+ levels may have dropped 50% or more from youthful levels. This means most adults over 40 can potentially benefit from NAD+ support regardless of whether they have symptoms.

Subcutaneous Injection

LOADING PROTOCOL
Dose: 100 to 200 mg daily
Duration: 7 to 10 consecutive days
Purpose: Rapidly restore depleted cellular reserves

MAINTENANCE PROTOCOL
Dose: 50 to 100 mg
Frequency: 1 to 3 times per week
Timing: Morning preferred (may interfere with sleep if taken late)

Cycling (Non-Negotiable for Ongoing Use)

Do 8 to 12 weeks on NAD+ supplementation, then 4 to 8 weeks off. This prevents NAMPT suppression and gives your body's natural production pathways a chance to recover. Exercise during the off-cycle helps restore natural NAD+ production.

Reconstitution

NAD+ requires more bacteriostatic water than most peptides. It is a large molecule that can crystallize or precipitate out of solution when the concentration is too high. The standard 2 mL reconstitution used for most peptides does not work here.

RECONSTITUTION
500 mg vial: Add 3 mL bacteriostatic water
Concentration: 166.7 mg/mL
For 100 mg dose, draw 60 units on insulin syringe

1000 mg vial: Add 6 mL bacteriostatic water
Concentration: 166.7 mg/mL
For 100 mg dose, draw 60 units on insulin syringe

NAD+ typically comes in 10 mL glass vials, so both volumes fit with room to spare. Let the water run down the inside wall of the vial. Do not spray directly on the powder. Swirl gently until fully dissolved. Do not shake.
After reconstitution, refrigerate at 2 to 8 degrees Celsius. Use within 14 days (not the typical 28 days for most peptides). Do not freeze. Protect from light. Before every injection, inspect the solution. It should be clear and colorless. If you see any cloudiness, discoloration, or particles, the NAD+ has precipitated out of solution and that vial should be discarded.

Stacking Context

The Cellular Energy Protocol

This is the framework that ties all the mitochondrial compounds together. It happens in two phases.
Phase 1: Repair (Weeks 1 through 8) SS-31 to repair mitochondrial membrane structure. NAD+ (injectable or niacin) to provide substrate through the biosynthesis pathways. 5-amino-1MQ to block the drainage of NAD+ building blocks in the salvage pathway. Creatine to support the overall ATP system.
Phase 2: Optimize (Weeks 5 through 12) Replace SS-31 with MOTS-c, which helps your body optimize how efficiently the repaired mitochondria run. Continue NAD+ and creatine. Continue 5-amino-1MQ if desired.

Common Questions

What is the difference between NAD+, NMN, and NR? NAD+ is the active molecule your cells use. NMN and NR are precursors, meaning your body converts them into NAD+. NMN is one step away from becoming NAD+. NR converts to NMN inside your cells, then becomes NAD+. Oral NAD+ itself is poorly absorbed, which is why precursors or injectable forms are used.
Are injections better than oral supplements? Injections provide high bioavailability directly into the bloodstream and bypass the gut conversion problem. Whether this translates to greater intracellular NAD+ than oral precursors is still debated. Some experts believe oral precursors may actually be more effective at raising NAD+ inside cells because they enter cells before conversion.
Why not just take niacin? If you can tolerate the flush, niacin is actually your cheapest and most proven option. The flush diminishes over time with consistent use. The main reason people choose NMN, NR, or injectable NAD+ is to avoid that flush.
Why do I need to cycle NAD+? Chronic supplementation can suppress NAMPT, the key enzyme in the salvage pathway that produces about 80% of your NAD+. If you suppress it long enough and then stop supplementing, your body may produce less NAD+ naturally than before you started. Cycling (8 to 12 weeks on, 4 to 8 weeks off) prevents this.
Can I take NAD+ if my mitochondria are damaged? You can, but you may not get much benefit from it alone. Converting NAD+ precursors into usable NAD+ requires ATP, and damaged mitochondria cannot produce ATP efficiently. Consider using SS-31 first to repair the structural damage, then add NAD+ once the machinery can actually use it.
When should I take it? Morning is generally preferred because NAD+ boosts cellular energy production. Taking it later in the day may interfere with sleep in

some individuals.

Side Effects

In Published Research

Oral NAD+ precursors (NMN, NR) have been studied at doses up to 1000 to 2000 mg daily and appear safe in the short to medium term. No significant toxicity has been reported at standard dose

- Flushing and warmth (primarily with niacin)
- Nausea
- Headache
- Fatigue
- Stomach discomfort
- Injection site reactions (redness, tenderness, stinging)

Important notes:

- Side effects are often dose and rate dependent
- Starting with lower doses helps assess tolerance
- Most side effects resolve quickly
- Long-term safety data beyond 12 months is still limited

What Users Report

Injectable NAD+ is commonly described as more uncomfortable to inject than other peptides. A stinging or burning sensation at the injection site is frequently reported. Injecting slowly helps. Some users report mild nausea, especially at higher doses.

Storage and Handling

Injectable NAD+ (After Reconstitution)

Refrigerate at 2 to 8 degrees Celsius. Do not freeze. Use within 14 days (shorter stability window than most peptides due to the molecule's size and tendency to precipitate). Protect from light. Inspect for clarity before each use. If cloudy, discolored, or if you see particles, the NAD+ has crashed out of solution. Discard the vial.

Lyophilized Powder (Before Reconstitution)

Store at room temperature or refrigerated. Keep dry and away from light. Stable for much longer in lyophilized form.